
What are Marine Layer Clouds and How Do they Form?

- [What are Marine Layer Clouds?](#)
 - [Mixing Air and the Lifting Condensation Level](#)
 - [The Pacific High and Sinking Air](#)
 - [The Inversion Layer](#)
 - [Position of the LCL and Inversion Base](#)
 - [Advection of Marine Layer Clouds](#)
 - [Inland Extent of Marine Layer Clouds](#)
 - [Why Marine Layers Do Not Form over All Water Bodies?](#)
 - [How Do Marine Layer Clouds Dissipate?](#)
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Marine Layer Clouds

Marine Layer clouds that impact California are low altitude stratus clouds that form over the adjacent ocean waters. Once formed, they may be advected by the wind over land areas. Stratus type clouds are sheet like clouds with close to horizontally uniform base and top. They generally extend for large distances horizontally (10-100s of kms), but are relatively shallow in depth usually (usually 500-2000 meters).

Mixing Air and the Lifting Condensation Level

The formation of these clouds usually begins when wind over the water surface mixes moist surface air upwards. As this air moves up, it expands and cools. The cooling causes the relative humidity to increase and once the relative humidity reaches 100%, condensation of water vapor into liquid water drops takes place and clouds begin form. The depth through which the air mixes is referred to as the *mixed layer*.

The altitude (or height) that surface air must rise for condensation to start is called the *Lifting Condensation Level* or LCL for short. So if air near the surface is allowed to mix upwards to the LCL a cloud will form. Under certain conditions air does not mix high enough for clouds to form, that is the top of the mixed layer is beneath the LCL. While no clouds formed, the mixing has increased the relative humidity near the top of the mixed layer and subsequent cooling of this air could increase the relative humidity to 100% and initiate cloud formation.

The height of the LCL depends on several variables including the water temperature and the temperature and moisture content of the air above the water.

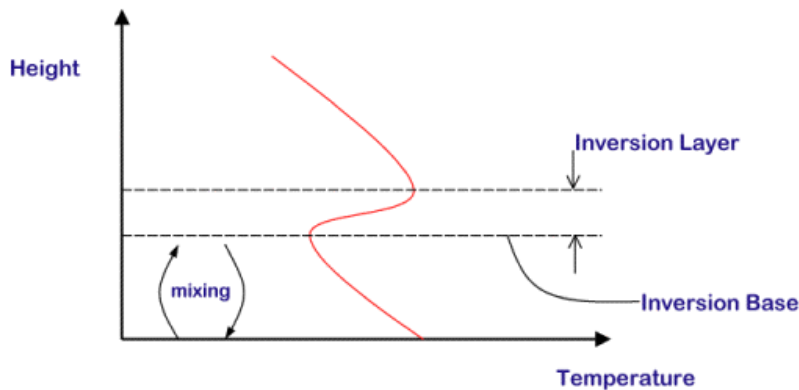
The Pacific High and Sinking Air

The Pacific High is a large region of high pressure that is usually present over Southern California and the adjacent ocean. Within the Pacific High air is sinking towards the surface, often at rates more than 1 km per day. The sinking causes the air to compress and warm. So, over the ocean off of Southern California there are two competing processes:

- Mixing of surface air ==> leads to rising air that cools
- the Pacific High ==> leads to sinking air that warms

Often, the net result of these two processes is an *Inversion Layer*.

The Inversion Layer



Usually, temperature decreases as one moves upward in the atmosphere. However, in an inversion layer the temperature increases with height and forms a very stable layer that acts as a lid keeping the air beneath from penetrating higher into the atmosphere. As seen in the figure above, the *inversion base* limits the vertical extent of air mixing near the surface. This type of inversion is often called a *subsidence inversion* or a *marine air inversion*.

The strength of the inversion is often measured by the temperature difference between the top and base of the inversion layer, with a larger temperature difference indicating a stronger inversion. Stronger (weaker) than normal high pressure is usually associated with a stronger (weaker) than normal inversion layer. Stronger than normal inversion layers will often result in more marine layer cloud formation and longer times for the clouds to dissipate. The strength of the inversion can also be modulated by climate features such as El Niño and the Pacific Decadal Oscillation (PDO).

The development of the inversion layer is enhanced by the relatively cool ocean water off of California which helps increase the contrast between the cold air below the inversion layer and the warm air above the inversion layer. The California current system flows from Alaska southward to California bringing relatively cool water from the polar regions. Upwelling of cold deeper water along the coast of California also helps to keep the surface waters cool. Colder than normal ocean water often leads to stronger than normal inversions that can increase the amount and duration of marine layer clouds.

Another type of inversion often seen in California is called a *radiation inversion*. Radiation inversions form when the ground cools during the night resulting in an inversion that is based at the surface, that is the temperature increases as one moves up from the surface. Radiation inversions are usually strongest in the early morning hours during the winter months. Unlike subsidence inversions which can persist for long periods of time, a radiation inversion usually dissipates a few hours after sunrise.

Although radiation inversions generally do not play a major role in the formation or dissipation of marine layer clouds, they (along with subsidence inversions) have an impact on pollution levels by inhibiting vertical mixing of air near the surface.

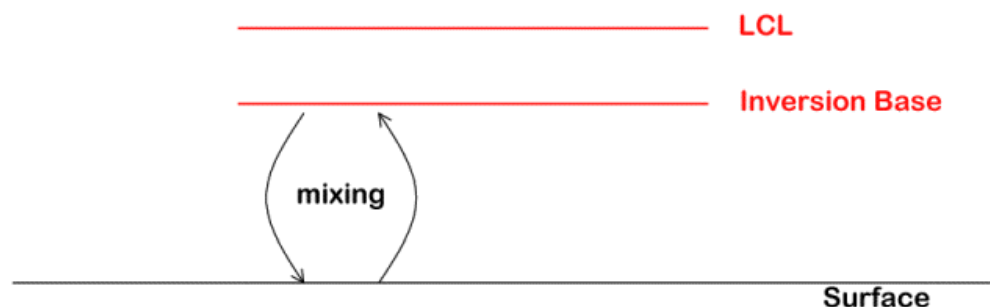
LCL vs. Inversion Base

The following diagrams illustrate how the elevation of the inversion layer base can impact the formation and subsequent duration of marine layer clouds. For this illustration we are assuming idealized conditions, that is that the air between the surface and inversion base is perfectly mixed. We are also assuming that the elevation of the inversion base and LCL is horizontally uniform.

The relative location of the LCL and the inversion base determine if marine stratus clouds will form.

Case A: LCL Above Inversion Base

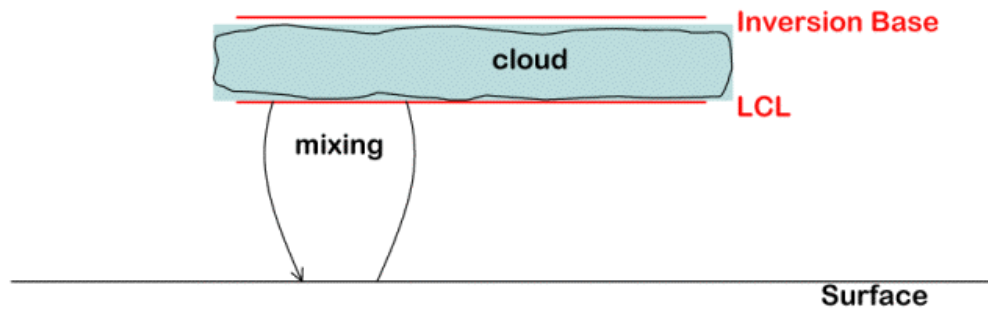
In this case the inversion base prevents mixing from reaching the LCL. The relative humidity does not reach 100% and no clouds will form. This situation occurs when there is a very strong high pressure over the region and/or very weak mixing (low winds).



Case B: LCL Below Inversion Base

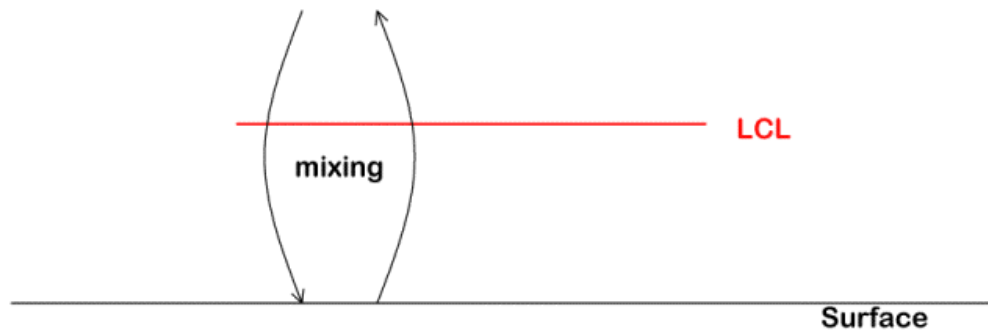
Here, mixing is allowed to extend above and beyond the LCL up until the Inversion Base is encountered. In this case a cloud forms between the LCL and Inversion Base. This situation is typical when there is a moderate to strong high pressure over the region and moderate to strong mixing. This is how marine layer clouds are formed. The base of the inversion layer limits the vertical growth of clouds and is why the top of marine layer clouds are usually very uniform in height. Airplanes flying out of Lindbergh Field during summer

mornings often will travel through the marine layer a short time after takeoff. Once the airplane emerges through the top of the clouds, passengers experience bright sunshine and can see the very sharply delineated top of the marine layer.



Case C: No Inversion Layer (or Inversion Layer at very high altitude)

Without an inversion layer present (or if the inversion layer is present but at a very high height far above the LCL), mixing will extend far above the LCL. As the surface air moves higher it can mix with much drier air from above. If too much dry air is mixed in, the relative humidity will fall below 100% and any clouds that formed will evaporate.



It is the relative location of both the LCL and the base of the Inversion Layer that determine if marine layer clouds will form. The most likely time of the year for marine layer clouds to form is April to August. The local winds are most likely to bring the marine stratus over the coastal land regions during May-June. However, winds can (and do) bring marine layer clouds over land during other times of the year as well.

Advection of Marine Layer Clouds

Once the marine stratus clouds form over the water they can be advected over coastal land areas by the winds. The prevailing winds in Southern California during the summer months are generally westerly (air moving from the west to the east) which helps bring these clouds onto the coastal areas. Additionally, a sea-breeze circulation often develops during the warm summer days which can increase the onshore flow of air helping maintain a steady stream of marine stratus clouds.

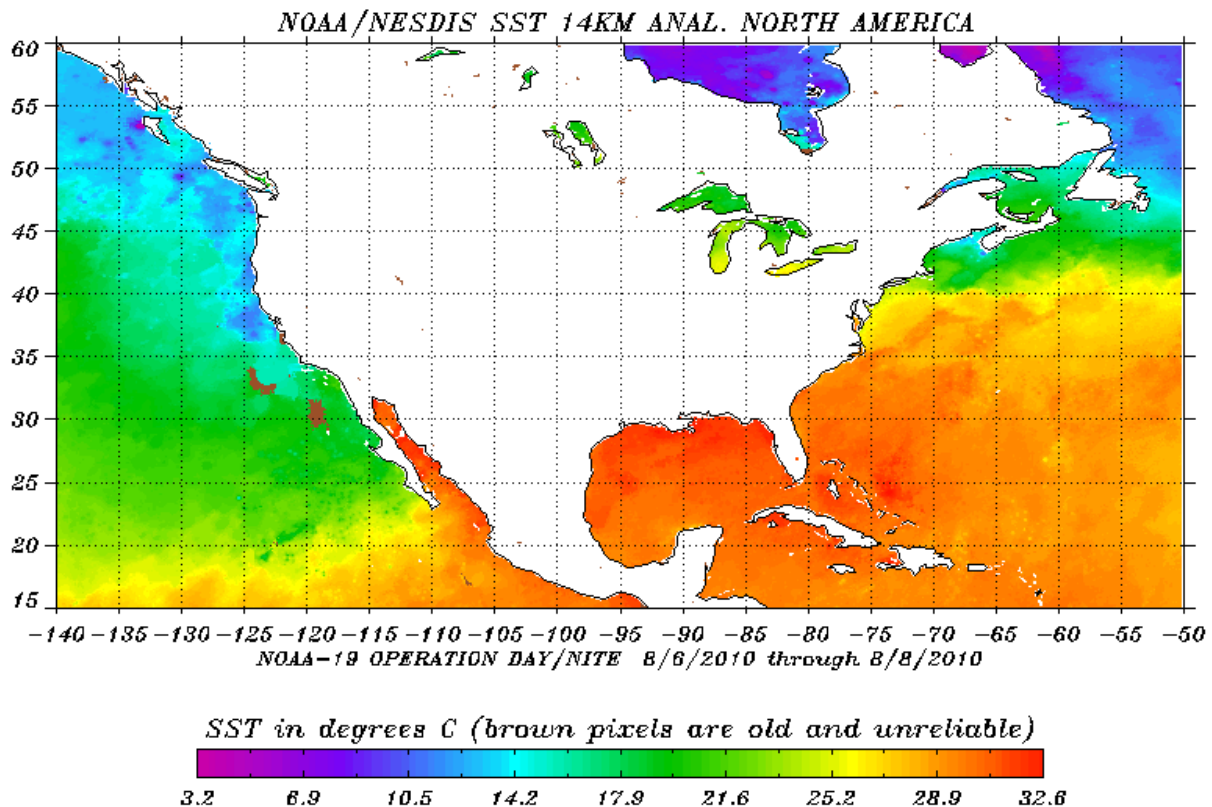
Inland Extent of Marine Layer Clouds

The horizontal extent of these clouds is limited by the hills and mountains throughout Southern California. Onshore winds will continue to push the clouds inland until they encounter land that is at the same elevation as the clouds themselves. The hills and mountains act as a barrier to any further inland extent of these clouds. This can often result in very low clouds and even fog (fog is defined as a cloud in contact with the ground) in the foothills throughout the region. Locations with an elevation higher than the clouds tops can provide some spectacular views of the clouds covering the low-lying valleys to the west.

Why Marine Layer Clouds Do Not Form over all Large Bodies of Water

For marine layer clouds to form a strong inversion layer must be present. One of the main ingredients for an inversion layer is the sinking air found within high pressure systems. Large semi-permanent high pressure systems such as the Pacific High are only found at certain locations throughout the world which limits the formation of persistent inversion layers to these regions.

Another factor helping inversion layers form is relatively cold water temperatures. The cold water cools the air beneath the inversion layer that helps provide the strong contrast with the warmer air above the inversion layer.



The above map shows the sea surface temperature (SST) off the coasts of the United States for a period during August 2010. Not all areas beneath the large high pressure systems such as the Pacific High have water cold enough to form an inversion layer. For instance, the south east coast of the United States is often under the Bermuda High pressure system, but the water temperatures are usually about 11°C (20°F) warmer than the water off of California. With such warm water it is more difficult for an inversion layer to form.

How Do Marine Layer Clouds Dissipate?

Marine layer clouds often reach their maximum extent around sunrise as this is when the air near the surface usually reaches a minimum temperature. The colder surface temperatures enhance the inversion layer and also make it more likely the air beneath the inversion base is saturated with a relative humidity of 100%.

As the day progresses, sunlight that penetrates through the clouds will warm the surface and the air above. The warming is greatest over land areas as land heats up much faster than water. As the air warms it is mixed upwards and will begin to mix into the clouds. This warming of the cloudy air decreases the relative humidity below the 100% level and the cloud begins to evaporate. Strong winds above the clouds can mix in drier air also leading to more evaporation. A thicker marine layer will dissipate slower as it will take more time for enough warm and/or drier air to mix in and evaporate all the cloud.

[Back to Main Marine Layer Page](#)